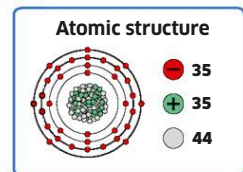


Halogens and noble gases

On the right-hand side of the periodic table are the non-metals known as halogens (group 17) and noble gases (group 18).

The word “halogen” means “salt-forming,” and refers to the fact that these elements easily form salt compounds with metals. These include sodium chloride—common table salt—and those metal salts that give fireworks their colors, such as barium chloride which makes green stars. The noble gases don’t form bonds with other “common” elements and are always gases at room temperature.



BROMINE

Bromum

Discovered: 1826

One of only two elements that are liquids at room temperature (the other is mercury), bromine is toxic and corrosive. It forms less harmful salt compounds, such as those found in the Dead Sea in the Middle East.

A liquid halogen

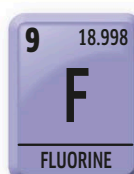
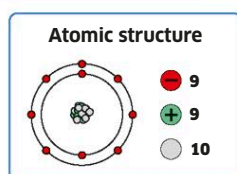
A drop of pure, dark orange-brown bromine fills the rest of the glass sphere with paler vapor.

FLUORINE

Fluor

Discovered: 1886

A pale yellow gas, fluorine is an incredibly reactive element. On its own, it is very toxic, and ready to combine with even some of the least reactive elements. It will burn through materials such as glass and steel. When added to drinking water and toothpaste in small doses, it helps prevent tooth decay.



Calming mixture

In this glass vial, fluorine has been mixed with the noble gas helium to keep it from reacting violently.

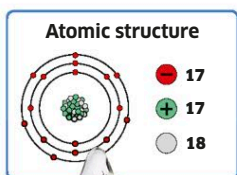


CHLORINE

Chlorum

Discovered: 1774

Like its periodic neighbor fluorine, chlorine is a very reactive gas. It is so poisonous that it has been used in chemical warfare, in World War I for example. It affects the lungs, producing a horrible, choking effect. Its deadly properties have been put to better use in the fight against typhoid and cholera: when added to water supplies, it kills the bacteria that cause these diseases. It is also used to keep swimming pools clean, and in household bleach.



Chlorine is a pale green gas.

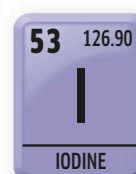
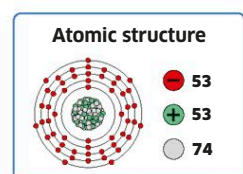


IODINE

Iodium

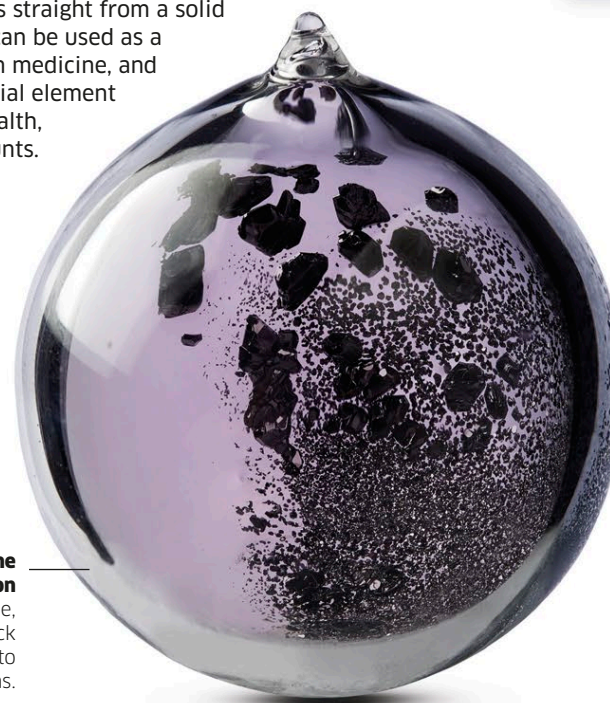
Discovered: 1811

The only halogen that is solid at room temperature, iodine will sublime, which means it turns straight from a solid into a gas. It can be used as a disinfectant in medicine, and it is an essential element for human health, in small amounts.



Iodine sublimation

The dark purple, almost black solid turns into a paler gas.



TENNESSINE

Tennessine

Discovered: 2010

A latecomer among the halogens, this artificial element only got its name in 2016, six years after being created. It doesn't exist naturally, but is produced, a few atoms at a time, by crashing smaller atoms into one another until they stick together. The element is so new that so far, almost nothing is known about its chemistry. It is named after the US state of Tennessee, home to the laboratory where much of the research into making it took place.

